

Mark Shperkin

Applied AI Engineer

Contact Info

- ✉ markshperkin1@gmail.com
- 🌐 [linkedin.com/in/mark-shperkin](https://www.linkedin.com/in/mark-shperkin)
- 🐙 github.com/markshperkin
- 📞 +972-544386995

About Me

Applied AI Engineer building production-grade AI applications using LLMs, RAG, embeddings, and AI agents.

B.S. and M.S. in Computer Science with hands-on experience designing and deploying scalable ML systems, retrieval pipelines, and automation workflows.

Experienced in prompt engineering, API integration, and end-to-end AI development from concept to production.

Fluent in Hebrew, English, and Russian.

Skills

Python, PyTorch, FastAPI, GenAI Applications, RAG, AI Agents, Embeddings, Vector Databases, Prompt Engineering, Context Engineering, LLM Evaluation, LangChain, API Integration, Software Architecture, System Design, OOP/OOD, Distributed Systems, Docker, Kubernetes, GitHub Actions (CI/CD)

Education

- 2024
2025
M.S. Computer Science
University of South Carolina
Post-Baccalaureate Certificate in Artificial Intelligence
(GPA 87)
- 2020
2024
B.S. Computer Science
University of South Carolina
(GPA 87, Student-Athlete)

Work Experience

Applied AI Engineer
RGIS

Israel | 2026

- Improved actionable lead quality and volume by 60% while reducing LLM cost and latency by developing a GenAI-driven lead intelligence platform with an agentic pipeline integrating multiple APIs for data retrieval, enrichment, and validation.
- Reduced manual research time from days to ~15 minutes by engineering LLM-based workflows for data aggregation, reasoning, scoring, and validation, implementing iterative evaluation loops and AI agent pipelines to identify high-value stakeholders for sales outreach.

Software Engineer
CleanRate (Startup)

Columbia, SC | 2024 – 2025

- Reduced energy-efficiency assessment turnaround from manual specialist reviews to near real-time evaluation by architecting HPXML generation pipelines and automated scoring systems, enabling scalable analysis of energy performance.

Teacher Assistant
University of South Carolina

Columbia, SC | 2024 – 2025

- Courses Taught: AI; Algorithmic Design II; Programming Languages & Structures
- Reduced manual grading effort by over 90% for 200+ weekly student submissions by developing Java/Python modules and automating evaluation workflows.

Project Experience

MarkGPT – Personal RAG/LLM Platform

2026

Frontend, Backend, Demo

- Built and deployed a production-grade RAG platform that enabled low-latency semantic retrieval and streaming AI responses by architecting an end-to-end inference pipeline using FastAPI, ChromaDB vector search, retrieval grounding, scope-restricted prompting, and Anthropic Claude.
- Deployed containerized AI services using Docker, CI/CD pipelines, and cloud-hosted infrastructure, enabling reliable end-to-end delivery of production LLM applications.

Master’s Thesis – Multi-Task Deep Learning with Single IMU

2025

ML Pipeline, Labeling Software (Demo), Publication

- Achieved ~0.77 micro-F1 on temporal activity segmentation and classification by designing and implementing an end-to-end PyTorch ML pipeline spanning data collection, labeling, model development, training, and evaluation.
- Designed and implemented anchor-based temporal detection algorithms, including multi-scale window generation, IoU-based anchor-ground-truth assignment, hard negative mining for class imbalance, and a two-stage 1D non-maximum suppression pipeline to produce precise, non-overlapping activity segments.

Other Experience

Professional Athlete – Swimming

2004 – 2024

Israeli National Team & University of South Carolina Gamecocks

- Represented the Israeli National Team and USC Division I SEC team, demonstrating elite discipline, teamwork, and high-performance consistency while balancing international competition, military service, and full-time academic commitments.

Military Service – First Sergeant
Israel Defense Forces – Air Force

Israel | 2016 – 2019

- Achieved ‘Elite Athlete’ status while balancing international competition with military responsibilities, earning an honorable discharge with a proven record of leadership, reliability, and high-pressure performance.